

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) v2.0

AI vs Human Content Detection System

Project Name: AI vs Human Content Detection System
Document Type: Functional Requirements Document (FRD)
Version: 2.0 Final
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Status: Approved for Client Handover
Date: June 2026

DOCUMENT CONTROL

Version Date		Author	Description
1.0	Project Initiation	Data Science Team	Initial FRD
2.0	June 2026	M. Sagar Reddy	Final Implemented Solution

APPROVAL MATRIX

Role	Responsibility	Status
Client Stakeholder	Functional Approval	Pending
Data Science Lead	Technical Approval	Completed
Client Deployment Team	Integration Approval	Pending

1. PURPOSE

This document defines the final functional requirements, system behavior, processing logic, prediction workflow, input/output specifications, and integration requirements for the AI vs Human Content Detection System.

The purpose of the solution is to automatically classify textual content as either:

- AI Generated
- Human Written

using a machine learning pipeline optimized for real-world content moderation scenarios.

2. SYSTEM OVERVIEW

The system accepts textual content as input and generates classification results indicating whether the submitted content is AI-generated or human-authored.

The final solution operates as a serialized machine learning pipeline and can be integrated into client-owned deployment infrastructure.

3. HIGH LEVEL FUNCTIONAL FLOW

Content Submission



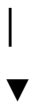
Input Validation



Text Cleaning



TF-IDF Feature Extraction



Logistic Regression Prediction



Probability Generation



Threshold Evaluation (0.44)



Final Classification

4. FUNCTIONAL REQUIREMENTS

FR-01 Content Input

Description

The system shall accept textual content for classification.

Required Input

Field	Required
-------	----------

text_content	Yes
--------------	-----

Example

Artificial Intelligence is transforming industries...

FR-02 Input Validation

Description

The system shall validate incoming content before prediction.

Validation Rules

Rule	Requirement
------	-------------

Empty Text	Reject
------------	--------

Null Text	Reject
-----------	--------

Non-String Content	Reject
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Expected Outcome

Invalid content shall not proceed to prediction.

FR-03 Text Cleaning

Description

The system shall standardize incoming text before feature extraction.

Processing Steps

- 1. Convert to lowercase
- 2. Remove special characters
- 3. Remove punctuation
- 4. Normalize whitespace

Output

Original:

"This Study Investigates AI Systems!"

Cleaned:

"this study investigates ai systems"

FR-04 Feature Extraction

Description

The system shall transform cleaned text into machine-readable numerical features.

Word Features

Component Value

Vectorizer	TF-IDF
N-Grams	(1,2)
Features	5,000

Character Features

Component Value

Vectorizer	TF-IDF
Analyzer	char_wb
N-Grams	(3,5)
Features	10,000

FR-05 Classification Engine

Description

The system shall generate prediction probabilities using Logistic Regression.

Model Configuration

Component Value

Algorithm	Logistic Regression
Class Weight	Balanced
Max Iterations	3000

FR-06 Probability Generation

Description

The system shall generate probability scores for both classes.

Output

{

```
"AI Probability": 0.91,  
"Human Probability": 0.09  
}
```

FR-07 Threshold Evaluation

Description

The system shall apply an optimized threshold before generating the final prediction.

Business Rule

AI_THRESHOLD = 0.44

Decision Logic

if ai_probability >= 0.44:

 Prediction = AI

else:

 Prediction = Human

FR-08 Final Prediction

Description

The system shall return a final classification label.

Supported Outputs

AI Generated

Human Written

5. OUTPUT REQUIREMENTS

Prediction Response Structure

```
{  
  "Prediction":  
    "AI Generated",  
  
  "AI Probability":  
    0.9124,  
  
  "Human Probability":
```

0.0876

}

6. DEPLOYMENT ARTIFACT REQUIREMENTS

The following files shall be generated and delivered.

File	Purpose
final_ai_detector.pkl	Serialized Model
decision_threshold.pkl	Threshold Logic
model_metadata.pkl	Model Metadata

7. MODEL METADATA REQUIREMENTS

The metadata file shall contain:

```
{  
  "model_name",  
  "accuracy",  
  "precision",  
  "recall",  
  "f1_score",  
  "threshold",  
  "training_samples",  
  "external_validation_samples"  
}
```

8. FUNCTIONAL BUSINESS RULES

Rule ID	Description
---------	-------------

FR-BR-01	Content must be cleaned before prediction
----------	---

FR-BR-02	TF-IDF features must be generated before scoring
----------	--

FR-BR-03	Threshold must be applied before classification
----------	---

FR-BR-04	Serialized artifacts must load successfully
----------	---

FR-BR-05	Prediction probabilities must be returned
----------	---

9. EXTERNAL VALIDATION SUPPORT

The system shall support evaluation on independent external datasets.

Validation outputs shall include:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

10. ERROR HANDLING REQUIREMENTS

Empty Text

Return:

Input text cannot be empty.

Invalid Data Type

Return:

Input must be string content.

Missing Model Artifact

Return:

Serialized model file not found.

11. PERFORMANCE REQUIREMENTS

Requirement	Target
Prediction Time	<500 ms
Classification Type	Binary
Supported Language	English
Deployment Format	Serialized Pipeline

12. ACCEPTANCE CRITERIA REVIEW

Requirement	Target	Actual
External Accuracy	≥85%	82.05%
Precision	≥90%	81.82%
Recall	≥80%	91.67%
F1 Score	≥85%	86.47%
Serialization	Required	Completed
External Validation	Required	Completed

13. CLIENT INTEGRATION REQUIREMENTS

Client deployment team shall:

1. Load serialized artifacts.
2. Load threshold file.
3. Apply threshold during inference.
4. Monitor model performance.
5. Schedule future retraining activities.

14. PROJECT CLOSURE STATEMENT

The functional requirements defined within this document have been successfully implemented, validated, and packaged for deployment.

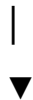
The final machine learning pipeline has been serialized and delivered to the client deployment team for integration into production environments.

APPENDIX A – DEPLOYMENT WORKFLOW

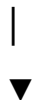
Load Model



Load Threshold



Receive Text



Clean Text



Predict Probability



Apply Threshold



Return Prediction
